



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
15-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TNAS]

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

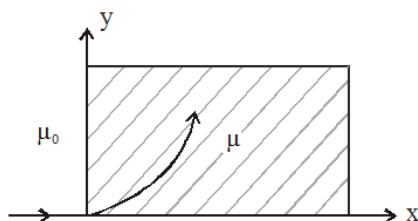
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

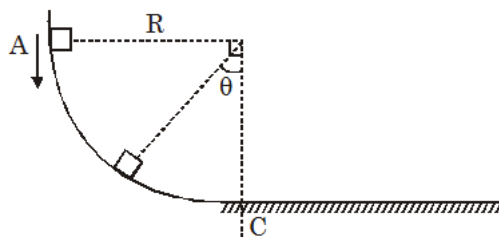
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. The refractive index of the medium with a certain region, $x > 0, y > 0$, changes with y . A thin light ray travelling in the x -direction in medium having refractive index $\mu_0 = 1$ strikes another medium of refractive index μ at right angles and moves through the medium along a circular arc of radius R as shown in the figure. The material with the greatest known refractive index is diamond, but even the refractive index of this material does not reach the value $\mu_{\max} = 2.5$. It is this limit that sets the maximum angular size of the arc the light ray can cover. Angular size of arc is the angle subtended by the arc at the centre.



- (A) The variation of refractive index μ with y is given as $\mu = \frac{R}{R-y}$
- (B) The unit vector in the direction of refracted light at $y = \frac{R}{2}$ is $\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}$
- (C) The maximum angular size of the arc of light is $\theta_{\max}$, the value of $\cos \theta_{\max}$ is $2/5$
- (D) The maximum angular size of the arc of light is $\theta_{\max}$, the value of $\cos \theta_{\max}$ is $3/5$

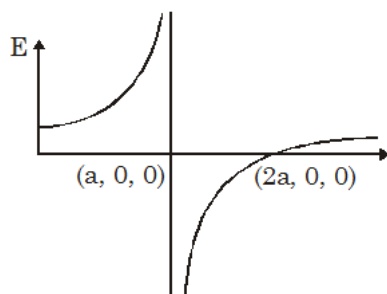
2. A particle of mass m is placed on vertical fixed rough circular track and then it is given velocity v vertically downward at position A on track. If block moves on track with constant speed then



- (A) Coefficient of friction between block and circular track as function of angle θ is $\mu = \frac{\sin \theta}{\cot \theta + \frac{v^2}{Rg}}$
- (B) Coefficient of friction between block and circular track as function of angle θ is $\mu = \frac{\sin \theta}{\cos \theta + \frac{v^2}{Rg}}$
- (C) Instantaneous power due to friction is $(-mg \sin \theta v)$
- (D) Work done from A to C by friction on block will be $(-mgR)$
3. A man is running with a constant acceleration (without slipping) on a plank which is placed on a horizontal smooth surface. Then choose the correct option :-
- (A) Work done by friction on the man is negative.
- (B) Work done by friction on the man is positive.
- (C) Work done by friction on the plank is positive.
- (D) Work done by friction on the plank is negative.

Space for Rough Work

4. The X-Y-Z space has an infinite charged sheet having surface charge density σ on the Y-Z plane. A negative charge particle $-Q$ is placed at $(a, 0, 0)$. The variation of net electric field on x-axis is shown in the figure. Then choose the correct option(s) :



- (A) $Q = \sigma 2\pi a^2$
- (B) $Q = \sigma 4\pi a^2$
- (C) The net electric field at the point $\left(\frac{a}{2}, 0, 0\right)$ is $\frac{5\sigma}{2\epsilon_0}$
- (D) At point $\left(\frac{5a}{2}, 0, 0\right)$ the net electric field is $\frac{5\sigma}{18\epsilon_0}$.

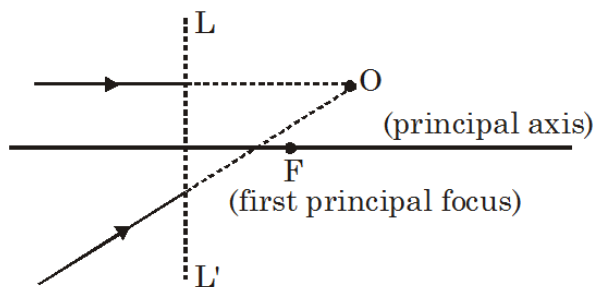
Space for Rough Work

5. A point charge q is kept at origin. The space around origin can be considered as non conducting sphere, tends to infinity and having continuous charge distribution, whose volume charge density is

$$\rho = \frac{-q}{4\pi a} \cdot \frac{e^{-\frac{r}{a}}}{r^2}$$

Where 'a' is a positive constant and 'r' is distance of a point from origin. The potential at infinity can be considered to be zero. The electric constant of space is 1. (Neglect the effect of polarization)

- (A) The total charge of system is zero.
- (B) The total charge of system is $\frac{q}{2}$.
- (C) The magnitude of electrostatic field at a distance of r from origin is $\frac{qe^{-\frac{r}{a}}}{4\pi\epsilon_0 r^2}$
- (D) The magnitude of electrostatic field at a distance of r from origin is $\frac{qe^{-\frac{r}{a}}}{8\pi\epsilon_0 r^2}$
6. Consider the rays shown in the diagram as paraxial. The image of the virtual point object O formed by the concave lens LL' is



- (A) Virtual. (B) Real.
- (C) Located below the principal axis. (D) Located left of the lens.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

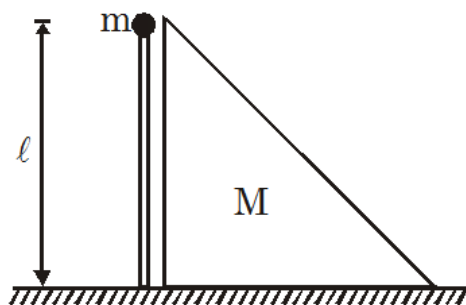
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

A weightless rod of length ℓ with a small load of mass m at one of its end is held vertical with its lower end hinged on a horizontal surface. The load touches a wedge of mass M in this position. A slight jerk towards right sets the system in motion (see figure), with rod rotating freely in vertical plane about its lower end. There is no friction.

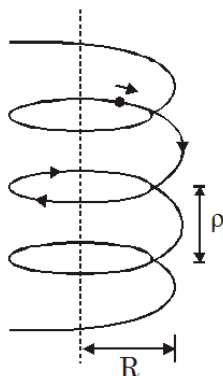


7. For what mass ratio $\frac{M}{m}$ will the rod form an angle $\theta = \frac{\pi}{3}$ with the vertical at the moment the load separates from the wedge ?
- (A) 1 (B) 2 (C) 3 (D) 4
8. What is speed of the wedge at that moment ? Neglect friction.
- (A) $\sqrt{\frac{g\ell}{2}}$ (B) $\sqrt{\frac{g\ell}{4}}$ (C) $\sqrt{\frac{g\ell}{6}}$ (D) $\sqrt{\frac{g\ell}{8}}$

Space for Rough Work

Paragraph for Questions 9 and 10

A long helix made of thin wire is held vertical. The radius and pitch of the helix are R and ρ respectively. A bead begins to slide down the helix. Pitch is defined as displacement of particle along the axis of helix in one rotation. (Given : $\tan \theta = \frac{\rho}{2\pi R}$)



9. Find the normal force applied by the wire on the bead when the speed of the bead is v .

(A) $mg \cos \theta \sqrt{1 + \left(\frac{v^2 \cos \theta}{Rg} \right)^2}$

(B) $mg \sin \theta \sqrt{1 + \left(\frac{v^2 \cos \theta}{Rg} \right)^2}$

(C) $mg \sin \theta \sqrt{1 + \left(\frac{v^2 \sin \theta}{Rg} \right)^2}$

(D) $mg \cos \theta \sqrt{1 + \left(\frac{v^2 \sin \theta}{Rg} \right)^2}$

Space for Rough Work

10. Eventually, the bead acquires a constant speed of v_0 . Find the coefficient of friction between the wire and the bead.

$$(A) \mu = \frac{2 \tan \theta}{\sqrt{1 + \left(\frac{v^2 \sin \theta}{Rg} \right)^2}}$$

$$(B) \mu = \frac{\tan \theta}{\sqrt{1 + \left(\frac{v^2 \cos \theta}{Rg} \right)^2}}$$

$$(C) \mu = \frac{\tan \theta}{2\sqrt{1 + \left(\frac{v^2 \cos \theta}{Rg} \right)^2}}$$

$$(D) \mu = \frac{\tan \theta}{2\sqrt{1 + \left(\frac{v^2 \sin \theta}{Rg} \right)^2}}$$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

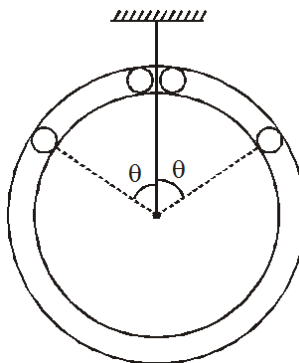
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A tube of mass M hangs from a thread and two balls of mass m slide inside it without friction (see figure). The balls are released simultaneously from the top of the ring and slide down on opposite sides. θ defines the positions of balls at any time as shown in figure.

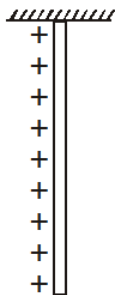


1. Minimum value of m so that ring will start to rise is kM , find value of k .
2. If $M = 0$, then the value of angle is θ at which the tube begins to rise. Find $\cos\theta$.

Space for Rough Work

Paragraph for Questions 3 and 4

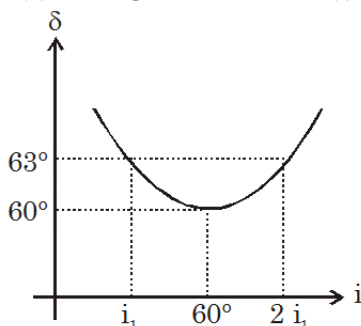
A uniformly charged non conducting rod is suspended vertically at its end. The rod can swing freely in the vertical plane without any friction. The linear charge density on the rod is λ C/m and it has a uniformly distributed mass of M . Length of the rod is L . A uniform horizontal electric field (E) is switched on in the region.



3. If value of electric field is E_0 so that the rod just manages to make itself horizontal. If $E_0 = \frac{nMg}{\lambda L}$, find n .
4. If electric field E_0 is switched on, the maximum angular speed acquired by the rod during its motion is $\sqrt{\frac{xg}{L}}$. Find x .

Paragraph for Questions 5 and 6

For a glass prism the plot of deviation (δ) v/s angle of incidence (i) is as shown.



5. Find the refractive index of the glass.
6. Find the value of angle i_1 (in degree).

Space for Rough Work

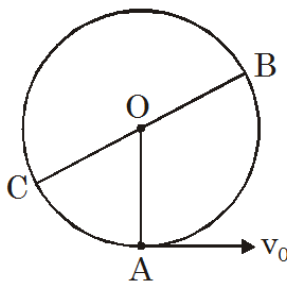
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

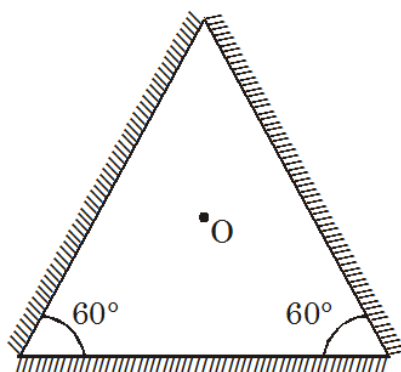
Zero Marks : 0 In all other cases

7. A heavy particle is attached to one end of a light string of length ℓ whose other end is fixed at O. The particle is projected horizontally with a velocity v_0 from its lowest position A. When the angular displacement of the string is more than 90° , the particle leaves the circular path at B. The string again becomes taut at C such that B, O, C are colinear. If $v_0^2 = \left(\frac{4 + n\sqrt{2}}{2} \right) g\ell$, find the value of n.

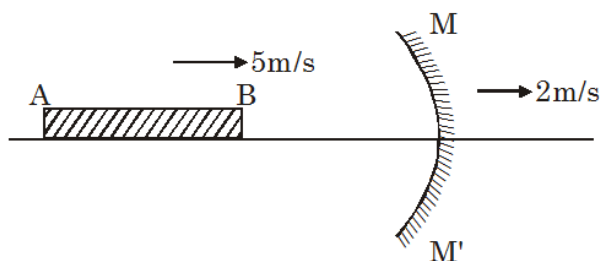


Space for Rough Work

8. Three plane mirrors are kept as shown in the figure. A point object (O) is kept at the centroid of the triangle seen in the figure. How many images will be formed ?



9. A pencil (AB) of length 20 cm is moving along the principal axis of a concave mirror MM', with a velocity 5 m/s approaching the mirror. The mirror itself is moving away from the pencil at a speed of 2 m/s. Radius of curvature of mirror is 40 cm. If the rate of change of length of the image of the pencil at the instant end A is at a distance of 60 cm from the mirror is $\frac{x}{4}$ m/s, find x.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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Negative Marks : -2 In all other cases.

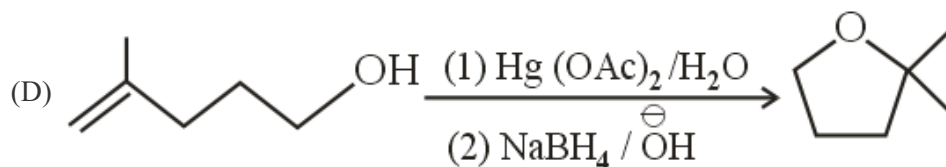
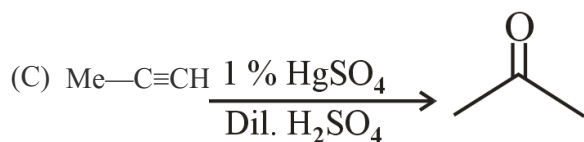
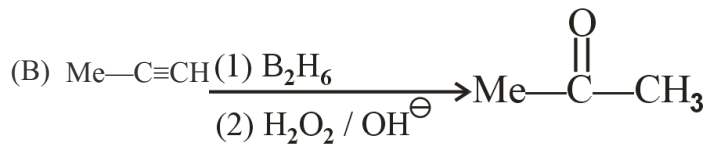
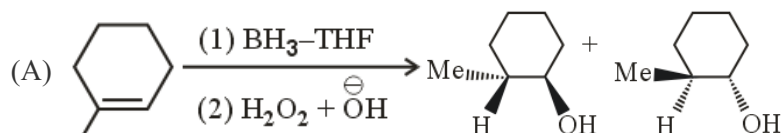
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following complex(es) may show secondary valency by Chloride/Bromide ion?



Space for Rough Work

2. Identify reactions correctly matched with their major products ?



3. Which of the following is/are bidentate ligand?

(A) en

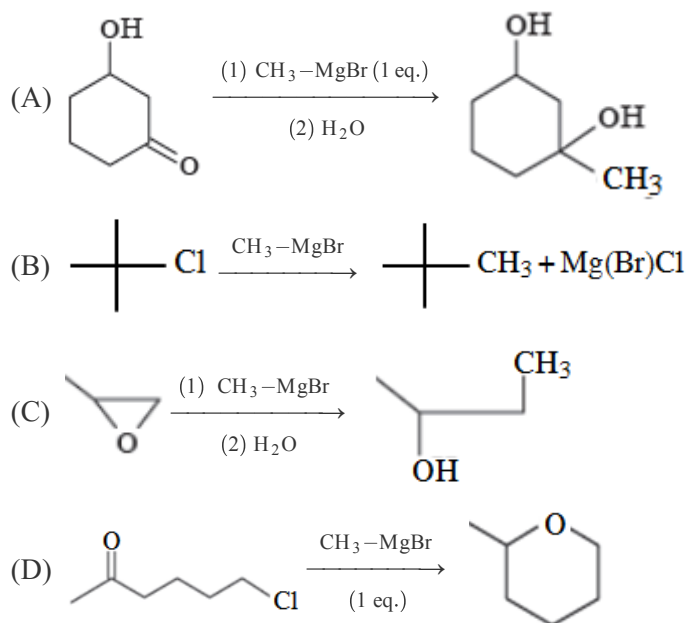
(B) dien

(C) acac^{-1}

(D) gly^{-1}

Space for Rough Work

4. Identify the reaction(s) with correct formation of major product ?

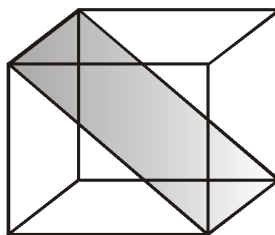


5. In which of the following pairs of aqueous solutions will osmotic pressure of solutions be almost equal?
[Assume 100% ionisation of salts]

- (A) 0.05 M $K_4[Fe(CN)_6]$ and 0.10 M $FeSO_4$
- (B) 0.05 M $K_4[Fe(CN)_6]$ and 0.05 M $FeSO_4(NH_4)_2 \cdot SO_4 \cdot 6H_2O$
- (C) 0.20 M $NaCl$ and 0.10 M $BaCl_2$
- (D) 0.05 M $FeSO_4(NH_4)_2SO_4 \cdot 6H_2O$ and 0.05 M $KCl \cdot MgCl_2 \cdot 6H_2O$

Space for Rough Work

6. Solid AB has ZnS-type structure with B^- ions at corners. If all ions presents in the plane as shown in figure are removed then empirical formula of final solid CANNOT be :



- (A) A_3B_2
 (B) A_4B_3
 (C) A_3B_4
 (D) A_4B_5

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
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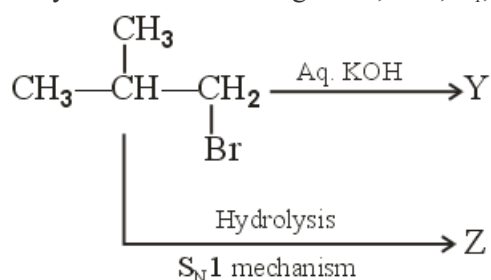
Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

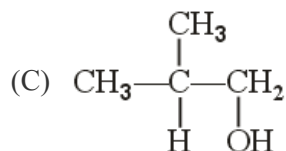
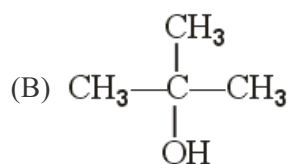
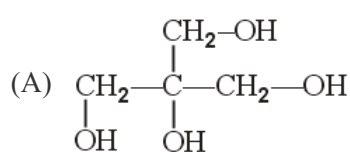
Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

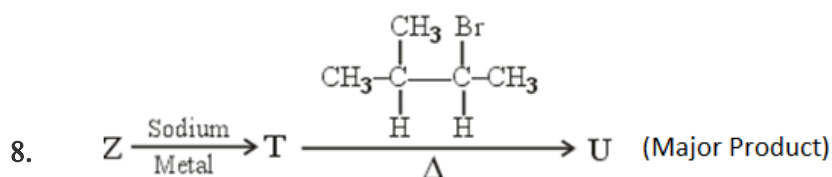
Alkyl halides can undergo SN^1 , SN^2 , E_1 , E_2 , E_{1cb} depending on various conditions.



7. Identify 'Y' :



(D) Both (B) & (C)



Select the correct statement :

- (A) Formation of U follows E^2 mechanism to give saytzeff product
- (B) Formation of U follows E_{1cb} mechanism to give saytzeff product
- (C) Formation of U follows E^1 mechanism to give Hoffmann product
- (D) Formation of U follows E^2 mechanism to give Hoffmann product

Space for Rough Work

Paragraph for Questions 9 and 10

An ideal solution is prepared at 25°C by dissolving 0.4 Kg of covalent solid AB (having rock-salt structure, B atoms at corner) and 5 moles liquid A in 20 moles liquid B. The solid is non-volatile and non-electrolyte in the solution. At 25° C, The vapour pressures of pure liquids A and B are 100 torr and 200 torr respectively and the vapour pressure of solution is 150 torr. The shortest distance between two nearest 'A' atoms in the solid is $200\sqrt{2}$ picometer. ($N_A = 6 \times 10^{23}$)

9. The molar mass of solid AB is -

(A) 100 g/mol

(B) 80 g/mol

(C) 40 g/mol

(D) 160 g/mol

10. The density of solid AB is -

(A) $\frac{25}{3}$ gm/cm³

(B) $\frac{50}{3}$ gm/cm³

(C) $\frac{25}{6}$ gm/cm³

(D) $\frac{25}{8}$ gm/cm³

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
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Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

The metal–carbon bond in metal carbonyls posses both σ & π character. The metal to ligand bonding creates a synergic effect which strengthens the bond between CO and the metal. Due to synergic effect M–C bond length & C–O bond length affected.

1. Consider $[\text{Co}(\text{CO})_4]^-$ as x and $[\text{Mn}(\text{CO})_6]^+$ as y then find how many statements are correct?
 - (i) The C – O bond length in x is more than that in y
 - (ii) The C – O bond energy in x is more than that in y
 - (iii) Double bond character in metal to carbon bond in x has more than that in y
 - (iv) Both x and y have the same EAN of the central metal atom

2. Consider the following species

$[\text{Ni}(\text{CO})_4]$, $[\text{V}(\text{CO})_6]^-$, CO^+ , CO_2 ,
 H_2CO_3 , CH_3COO^- , $[\text{NiP}(\text{CH}_3)_3(\text{CO})_3]$,
 $[\text{Cr}(\text{CO})_6]$, $[\text{NiPF}_3(\text{CO})_3]$

If bond length in CO is 'd' in isolated molecule.

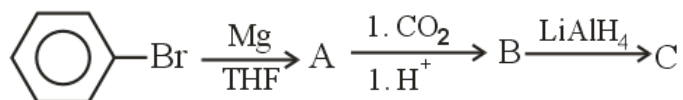
The number of species having higher C-O bond length then 'd' is/are 'X'

The number of species having lower C-O bond length then 'd' is/are 'y'

Find the value of $\left(\frac{X}{Y}\right)$

Space for Rough Work

Paragraph for Questions 3 and 4



3. D.O.U. of compound B = n_1 and D.O.U. of compound C = n_2 then value of $|n_1 - n_2|$
4. How many option(s) is/are correct
 - (i) (C) gives turbidity in 5 min with lucas reagent
 - (ii) (C) gives racemic mixture in product with PCl_3
 - (iii) (C) is isomers of anisole
 - (iv) (B) gives CO_2 gas with NaHCO_3

Paragraph for Questions 5 and 6

A mixture of H_2 , N_2 and O_2 occupying 100 ml underwent reaction so as to form H_2O_2 (l) and N_2H_2 (g) as the only products, causing the volume of contract by 60 ml. The remaining mixture was passed through pyragallol causing a contraction of 10 ml. To the remaining mixture excess H_2 was added and the above reaction was repeated, causing a reduction in volume of 10 ml. (No other products are formed)

5. What is the mole percent of N_2 in the initial mixture ?
6. What is the volume of percent of H_2 in the initial mixture ?

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Consider the following complex for given three statements :
 dichlorobis(ethylenediamine)chromium(III) chloride
 (i) Number of unpaired electrons in complex = (X)
 (ii) One mole of complex $\xrightarrow[\text{excess}]{\text{AgNO}_3}$ (Y) mole of AgCl.
 (iii) Total stereo isomeric forms of above complex = (Z) Find X + Y + Z
8. For the organometallic complex compound "Ferrocene" consider,
 x – Atomic Number of its central metal atom
 y – Oxidation number of its central metal atom
 z – Number of electrons provided by the ligand for sharing with its central metal atom
 Calculate x – y + z
9. The vapour pressures of two pure liquids A and B which form an ideal solution are 300 and 800 Torr, respectively, at temperature T. A liquid solution of A and B for which the mole fraction of A is 0.60 is contained in a cylinder closed by a piston on which the pressure can be varied. The solution is slowly vapourized at temperature T by decreasing the applied pressure. Find total vapour pressure (in torr) when last drop of liquid is remaining ?

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $(1 + \sqrt{2})^n = A_n + B_n \sqrt{2}$, where A_n and B_n are rational numbers and $n \in \mathbb{N}$, then the value of

$\lim_{n \rightarrow \infty} \frac{A_n}{B_n}$ is equal to -

(where $[.]$ denotes greatest integer function)

(A) $\left[\lim_{x \rightarrow 0} \frac{\sin x}{x} \right]$

(B) $\lim_{x \rightarrow 0} \left[\frac{\sin x}{x} \right]$

(C) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{|\cos x - \sin x|}{|x - \frac{\pi}{4}|}$

(D) $\left[\lim_{x \rightarrow 0} \frac{\sin \sqrt{2} x}{x} \right]$

Space for Rough Work

2. $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 - \frac{4x}{3}} + x + 2 \right)$ is equal to-

- (A) $\frac{4}{3}$ (B) $-\frac{4}{3}$ (C) $\frac{8}{3}$ (D) $-\frac{8}{3}$

3. Given $\lim_{x \rightarrow \alpha} \frac{\sin([x^3] - [x]^3)}{(x - \alpha)^3}$ exists finitely, then the value of α can be (where $\alpha \in \mathbb{I}$ & $[.]$ denotes greatest integer function)-

- (A) 0 (B) 1
(C) -1 (D) 2

4. Let $f(x) = \begin{cases} \min \{ \sin^{-1} x, \cos^{-1} x \} & ; \quad 0 \leq x \leq 1 \\ \max. \{ \sin^{-1} x, \cos^{-1} x \} & ; \quad -1 \leq x < 0 \end{cases}$ and $f(x) = \frac{1}{2}$, $f(x) = \frac{1}{4}$ & $f(x) = 2$ has number of solution(s) $\lambda_1, \lambda_2, \lambda_3$ respectively, then

- (A) $\lambda_1 = 2$ (B) $\lambda_2 = 1$
(C) $\lambda_2 = 2$ (D) $\lambda_3 = 1$

Space for Rough Work

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos^{-1}(-\{x\})$, where $\{x\}$ is fractional part function, then which of the following is/are true ?
- (A) $f(x)$ is many one function
- (B) period of $f(x)$ is 1
- (C) graph of $f(x)$ lies above x-axis
- (D) $f(x)$ is neither even nor odd function
6. If a function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $\left| \sum_{k=1}^n 3^k (f(x+ky) - f(x-ky)) \right| \leq 1$ for every $n \in \mathbb{N}$ and for all $x, y \in \mathbb{R}$, then which of the following is/are true
- (A) $|3^n (f(x+ny) - f(x-ny))| \leq 2$
- (B) $|f(u) - f(v)| \leq \frac{2}{3^n}$
- (C) $f(x)$ is a constant function
- (D) $f(x)$ is an odd function

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

If $\tan^2\alpha \tan^2\beta + \tan^2\beta \tan^2\gamma + \tan^2\gamma \tan^2\alpha + 2\tan^2\alpha \tan^2\beta \tan^2\gamma = 1$.

(where $\alpha, \beta, \gamma \in \mathbb{R} - \left\{ \frac{n\pi}{2} \right\}, n \in \mathbb{I}$)

7. Value of $(\prod \cot^2\alpha - \sum \cot^2\alpha)$ is equal to -

(A) 0 (B) -1 (C) 1 (D) 2

8. Value of $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$ is equal to -

(A) -1 (B) 0 (C) 1 (D) 2

Space for Rough Work

Paragraph for Questions 9 and 10

The sequence $\{a_n\}_{n=1}^{+\infty}$ is defined by $a_1 = 0$ and $a_{n+1} = a_n + 4n + 3$, $n \geq 1$

9. The remainder when a_{50} is divided by 50, is

(A) 0 (B) 3 (C) 29 (D) 47

10. The value of $\lim_{n \rightarrow \infty} \frac{\sqrt{a_n} + \sqrt{a_{4n}} + \sqrt{a_{4^2n}} + \sqrt{a_{4^3n}} + \dots + \sqrt{a_{4^{10}n}}}{\sqrt{a_n} + \sqrt{a_{2n}} + \sqrt{a_{2^2n}} + \sqrt{a_{2^3n}} + \dots + \sqrt{a_{2^{10}n}}}$ equals

(A) 685 (B) 683 (C) 684 (D) 680

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider $f(x) = \tan^{-1} \left(\frac{|x|}{\sqrt{1-x^2}} \right) + \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$ then

1. Number of integers in the range of $f(x)$ is
2. For $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2} \right)$, number of solution satisfying $f(\sin \theta) < f(\cos \theta)$ is

Space for Rough Work

Paragraph for Questions 3 and 4

Let $a = \sin 10^\circ$, $b = \sin 50^\circ$, $c = \sin 70^\circ$.

3. Value of $abc \left(\frac{a+b}{c} \right) \left(\frac{1}{a} + \frac{1}{b} - \frac{1}{c} \right)$ is equal to -
4. Value of $\frac{a^4 + b^4 + c^4}{(ab)^2 + (bc)^2 + (ca)^2}$ is -

Paragraph for Questions 5 and 6

Consider the function $f : A \rightarrow [0,5]$ $f(x) = \sqrt{9 - (x-1)(x-3)(x-5)(x-7)}$.

(Where set A represents all values of x for which f(x) is defined.)

5. In the domain of the function, number of possible integral values of x is equal to
6. Among the following, the number of correct statements is
 (i) function is many one (ii) function is one-one
 (iii) function is onto (iv) function is into

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. Let $f(x)$ be a function satisfying $f(x) + f(x-1) = x^2, \forall x \in \mathbb{R}$. If $f(19) = 94$ then the remainder when $f(94)$ is divided by 1000 is
8. Let $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$ where $a_i \in \mathbb{R}$ and $\lim_{x \rightarrow 0} \left(\frac{f(x) - x + 2x^2}{3x^3} \right)^{\frac{2}{x}} = e^4$, then $\sum_{i=0}^4 a_i$ is equal to
9. A function f satisfies $f(3x) = 3f(x) \forall x \in [1, \infty)$ and $f(x) = 1 - |x - 2|$ for $x \in [1, 3]$. The smallest x for which $f(x) = f(2001)$ is
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work